Peaceman–Rachford against LOD for a diffusion step on a domain that changes every step

Development note (not for the paper)

1 Setting

The viscous step of FLIP solves, on the current set of fluid faces,

$$\partial_t u = \nu \nabla^2 u, \tag{1}$$

discretized in space as $u' = (A + B)\, u$, where A collects the x second differences and B the y ones. Let $k = \Delta t$ and

$$r = \frac{\nu k}{h^2} \tag{2}$$

be the diffusion number on a uniform grid of spacing h.

Unknowns. The vector u holds every face of the grid. A face that is not an unknown at step n (solid, air, pinned) keeps its value: its row of A_n and B_n is zero, and no unknown couples to it. So A_n and B_n act on one fixed space, and only their entries change with n.

Properties used. For each n:

(P1) On a staircase domain with shared interface viscosities, A_n and B_n are symmetric and negative semidefinite: $A_n = A_n^{\mathsf{T}}$, $x^{\mathsf{T}} A_n x \le 0$.

(P2) Their spectra lie in $[-4\nu/h^2,\, 0]$, the bound of the one-dimensional second difference.

(P3) In general $A_n B_n \neq B_n A_n$: the staircase breaks the tensor structure.

(P4) On a fixed domain $A_n = A$, $B_n = B$; on FLIP's moving free surface they change at every step.

2 The two schemes

Peaceman–Rachford (PR). Two half steps, each implicit in one direction and explicit in the other:

$$\left(I - \tfrac{k}{2} A_n\right) u^* = \left(I + \tfrac{k}{2} B_n\right) u^n, \tag{3}$$

$$\left(I - \tfrac{k}{2} B_n\right) u^{n+1} = \left(I + \tfrac{k}{2} A_n\right) u^*, \tag{4}$$

so $u^{n+1} = G_n u^n$ with

$$G_n = \left(I - \tfrac{k}{2} B_n\right)^{-1} \left(I + \tfrac{k}{2} A_n\right) \left(I - \tfrac{k}{2} A_n\right)^{-1} \left(I + \tfrac{k}{2} B_n\right). \tag{5}$$

Locally one-dimensional (LOD). Each sweep advances the full step in one direction, with no explicit operator:

$$u^{n+1} = L_n u^n, \qquad L_n = (I - kB_n)^{-1} (I - kA_n)^{-1}. \tag{6}$$

3 Two elementary lemmas

Lemma 1 (Cayley factor)**.** *Let M be symmetric negative semidefinite and $k > 0$. Then*

$$C = \left(I + \tfrac{k}{2}M\right)\left(I - \tfrac{k}{2}M\right)^{-1} \tag{7}$$

satisfies $\|C\|_2 \le 1$.

Proof. C is a function of the symmetric M, so it is symmetric with eigenvalues $(1+k\mu/2)/(1-k\mu/2)$, $\mu \le 0$ an eigenvalue of M. Each lies in $(-1, 1]$. □

Lemma 2 (Implicit factor)**.** *Let M be symmetric negative semidefinite. Then $\left\|(I - kM)^{-1}\right\|_2 \le 1$ for every $k \ge 0$.*

Proof. Its eigenvalues are $1/(1 - k\mu) \in (0, 1]$ for $\mu \le 0$. □

Remark. The explicit factor has no such bound. $I+\frac{k}{2}M$ has eigenvalues $1+k\mu/2$ with $\mu \in [-4\nu/h^2, 0]$ by (P2), so

$$\left\|I + \tfrac{k}{2}M\right\|_2 = \max(1,\ |1 - 2r|), \tag{8}$$

which exceeds 1 as soon as $r > 1$.

4 Fixed domain: PR is stable, without commutation

Proposition 1. *Let $A_n = A$ and $B_n = B$ for all n, both satisfying (P1)–(P2), and set $S = I - \frac{k}{2}B$. Then*

$$\|G^n\|_2 \le \kappa(S) \le 1 + 2r \qquad \textit{for all } n, \tag{9}$$

whether or not A and B commute.

Proof. $I + \frac{k}{2}B$ and S^{-1} are both functions of B, so they commute. From (5),

$$S\,G\,S^{-1} = \left(I + \tfrac{k}{2}A\right)\left(I - \tfrac{k}{2}A\right)^{-1}\left(I + \tfrac{k}{2}B\right)S^{-1} = C_A\,C_B, \tag{10}$$

with C_A, C_B the Cayley factors of Lemma 1. Hence $\left\|SGS^{-1}\right\|_2 \le 1$, and the similarity telescopes:

$$G^n = S^{-1}\,(C_A C_B)^n\,S \quad \Longrightarrow \quad \|G^n\|_2 \le \left\|S^{-1}\right\|_2 \|S\|_2 = \kappa(S). \tag{11}$$

By (P2) the eigenvalues of S lie in $[1,\ 1 + 2r]$, so $\kappa(S) \le 1 + 2r$. □

Remark. PR is a contraction in the norm $\|x\|_S = \|Sx\|_2$. In the plain norm a transient growth up to $1 + 2r$ is possible, but it never accumulates. Commutation of A and B is not needed; it would only make G itself a contraction in $\|\cdot\|_2$.

5 Moving domain: the similarity no longer cancels

Now A_n, B_n change at every step (P4). Let $S_n = I - \frac{k}{2}B_n$ and $C_n = C_{A_n}C_{B_n}$. By the same algebra, $G_n = S_n^{-1}C_nS_n$, and the product of N steps is

$$G_{N-1}\cdots G_1G_0 = S_{N-1}^{-1}\,C_{N-1}\left(S_{N-1}S_{N-2}^{-1}\right)C_{N-2}\cdots\left(S_1S_0^{-1}\right)C_0\,S_0. \quad (12)$$

The interior factors $S_jS_{j-1}^{-1}$ are the identity only when $B_j = B_{j-1}$. Since $\|C_j\|_2 \le 1$, and since $\|S_j\|_2\left\|S_{j-1}^{-1}\right\|_2 \le 1 + 2r$ for every j,

$$\|G_{N-1}\cdots G_0\|_2 \le (1+2r)^N. \quad (13)$$

Remark (What this does and does not prove). (13) is an upper bound: it shows that the argument guaranteeing PR's stability on a fixed domain fails on a moving one, and that each change of geometry can amplify by up to $1 + 2r$. It does not prove that growth occurs; that depends on how the domain changes. For $r \lesssim 1$ the bound is mild, and for $r \gg 1$, FLIP's viscous regime, it allows exponential growth. The mechanism is the explicit factor $I + \frac{k}{2}B_n$: its amplification, of norm up to $|1 - 2r|$, is undone only by the implicit factor $(I - \frac{k}{2}B_n)^{-1}$ with the *same* B_n, which the next step no longer applies.

6 LOD is a contraction for every sequence of domains

Proposition 2. *Under (P1), for any $k \ge 0$ and any sequence (A_n, B_n),*

$$\|L_{N-1}\cdots L_0\|_2 \le 1. \quad (14)$$

Proof. Each factor of (6) has $\|\cdot\|_2 \le 1$ by Lemma 2, and the norm is submultiplicative. No similarity transform is involved, so nothing depends on consecutive steps sharing an operator. □

7 Without symmetry: the maximum norm

FLIP's assembled viscous matrices are not exactly symmetric: the free-surface gating of a tangential coupling is decided from each row's own face, so a pair of rows can disagree. (P1) then fails and the 2-norm arguments above do not apply. A weaker structure survives: for each row i of $M = A_n$ or B_n,

$$m_{ii} = -d_i \le 0, \qquad m_{ij} \ge 0\ (j \ne i), \qquad \sum_{j\ne i} m_{ij} \le d_i, \quad (15)$$

that is, $-M$ is a weakly diagonally dominant M-matrix. A solid neighbour keeps its diagonal term and drops the coupling, which only strengthens the dominance.

Lemma 3. *Under (15), $\left\|(I - kM)^{-1}\right\|_\infty \le 1$ for every $k \ge 0$.*

Proof. $T = I - kM$ has diagonal $1 + kd_i > 0$, non-positive off-diagonal entries, and row sums $1 + k(d_i - \sum_{j\ne i} m_{ij}) \ge 1$. It is a strictly diagonally dominant M-matrix, so $T^{-1} \ge 0$ entrywise. With $\mathbf{1}$ the vector of ones, let $y = T^{-1}\mathbf{1} \ge 0$ (the row sums of T^{-1}) and $y_p = \max_i y_i$. Row p of $Ty = \mathbf{1}$ reads

$$1 = (1 + kd_p)\,y_p - k\sum_{j\ne p} m_{pj}\,y_j \;\ge\; (1 + kd_p)\,y_p - kd_p\,y_p = y_p. \quad (16)$$

So every row sum of the non-negative T^{-1} is at most 1, which is $\left\|T^{-1}\right\|_\infty \le 1$. □

Proposition 3. *Under* (15)*, LOD satisfies* $\|L_{N-1}\cdots L_0\|_\infty \le 1$ *for every* k *and every sequence of domains. The PR explicit factor, at an interior row with* $d_i = 2\nu/h^2$ *and* $\sum_{j\neq i} m_{ij} = d_i$*, has*

$$\left\|I + \tfrac{k}{2}M\right\|_\infty \;\ge\; |1-r| + r, \tag{17}$$

which is $2r-1>1$ *for* $r>1$*.*

Proof. The LOD bound follows from Lemma 3 and submultiplicativity. For the explicit factor, row i of $I+\frac{k}{2}M$ has diagonal $1-\frac{k}{2}d_i = 1-r$ and off-diagonal sum $\frac{k}{2}d_i = r$, all off-diagonal entries non-negative. □

Remark. Without symmetry there is also no norm in which each Cayley factor is a contraction for all n at once, so even on a fixed domain the PR argument of Proposition 1 is not available; for FLIP, which has both the moving domain and the asymmetry, nothing bounds PR, while LOD stays a contraction in $\|\cdot\|_\infty$.

8 Literature

Hundsdorfer and Verwer (1989) , *Math. Comp.* 53:81–101, Theorem 3.2. For dissipative, possibly non-commuting and time-dependent operators, they write PR with the explicit F_2 of step n evaluated at t_n, the same time level as the implicit F_2 that ended step $n-1$. Their transformation $\varepsilon_n \mapsto (I-\frac{\tau}{2}A_{2,n})\varepsilon_n$ is our S_n, and because both factors use the same $A_{2,n}$ the similarity telescopes exactly. This is Proposition 1 for time-dependent operators. The only loose end is the first step, whose transformed error $(I-\frac{\tau}{2}A_{2,0})\varepsilon_0$ "may become quite large and grow with spatial refinement". Their remedy is one backward Euler LOD step first. They also note that PR damps high-frequency errors only slowly, while LOD damps them strongly.

The link to FLIP. The free surface is re-marked between steps, so the operator of the explicit half step of step n is not the one of the implicit half step that ended step $n-1$. That is the factor $S_n S_{n-1}^{-1}$ of (12). In Hundsdorfer and Verwer's terms every FLIP step is a new first step, with a nonsmooth error, and their remedy applied every step is LOD throughout.

Related. Schatzman (1999), *J. Funct. Anal.* 162:219–255, shows that for non-commuting unbounded self-adjoint operators the PR product is not stable in general, the operator-level counterpart of $\kappa(S)\sim 1+2r$ growing without bound as $h\to 0$ (not read beyond the abstract). Li and Zhao (2017), *Appl. Math. Comput.* 299:28–44, report that PR loses unconditional stability at interfaces where Douglas ADI keeps it (not read).

9 Summary and predictions

	fixed domain	domain changing every step
PR, symmetric operators	bounded by $1+2r$	bound $(1+2r)^N$, growth possible for $r>1$
PR, non-symmetric (FLIP)	no bound available	no bound available
LOD, symmetric	$\|\cdot\|_2 \le 1$	$\|\cdot\|_2 \le 1$
LOD, M-matrix (FLIP)	$\|\cdot\|_\infty \le 1$	$\|\cdot\|_\infty \le 1$

Predictions to test on a 2D staircase diffusion problem, sweeping r from 0.1 to 100:

1. PR on a fixed staircase mask: $\|u^n\|$ bounded, possibly with transient growth up to $1+2r$.

2. PR on a mask that changes every step: growth possible once $r > 1$.

3. LOD: $\|u^n\|_2$ and $\|u^n\|_\infty$ non-increasing in every case.

The price of LOD is accuracy: its splitting error is first order in k, and by Weare (1979) any splitting carries the boundary error into the interior at a rate growing with the time step. For FLIP, whose other errors are first order already, stability is the property worth buying.